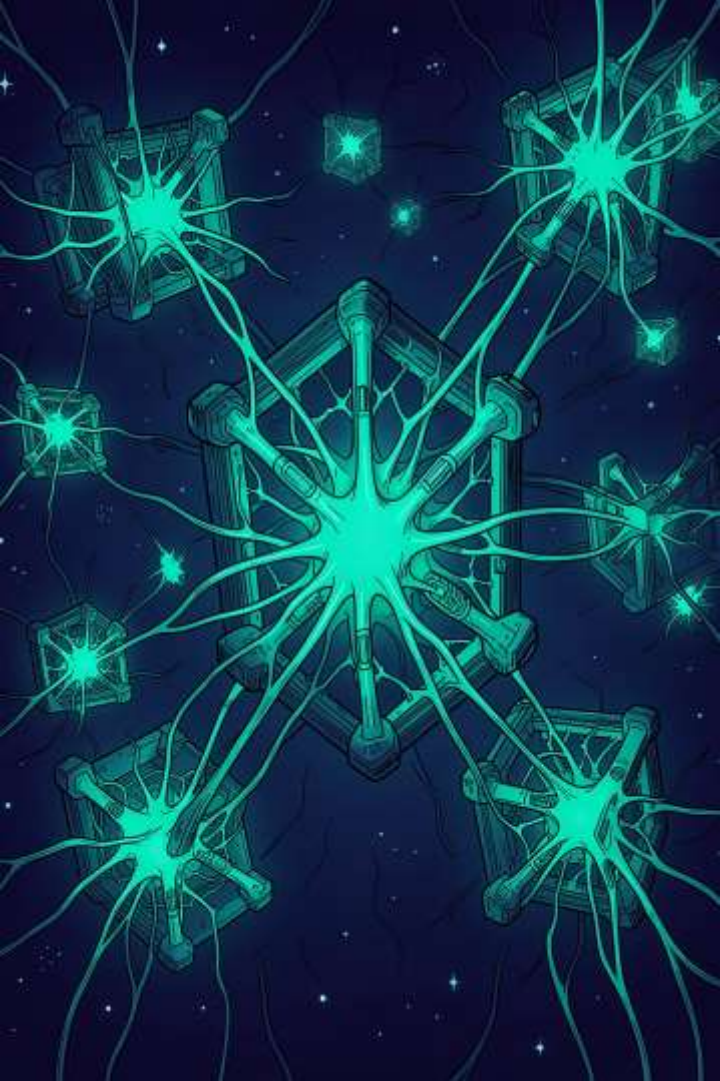




NLP SECURITY · RAG ROBUSTNESS

Retrieval Poisoning in RAG Systems

Adversarial semantic manipulation of dense retrieval — without prompt injection.

Team member:

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Attack Surface in RAG Pipelines

The adversary corrupts the retrieval corpus — not the LLM. Poisoned documents are semantically optimized to outrank clean results, injecting adversarial content before generation begins.



Query Input

User question enters the system.



Retriever Module

Dense retriever ranks candidate passages from the corpus.



Retrieved Chunks

Top-ranked context is passed forward for generation.



LLM Generation

The model conditions on retrieved context and the query.



Final Answer

System outputs the response.



Semantic spoof injection: adversarial documents are inserted into the corpus and influence the retriever *before* generation, rather than altering the prompt at runtime.

Dataset & Retrieval

SQuAD benchmark — ~5,000 clean documents
1,000 validation queries
MiniLM + FAISS / BM25 / Hybrid

Evaluation Metrics

Recall@5 · Recall@20
Top-1 Spoof Win Rate
Mean ± standard error reported

Contributions & Research Novelty

Semantic Retrieval Poisoning Without Prompt Injection

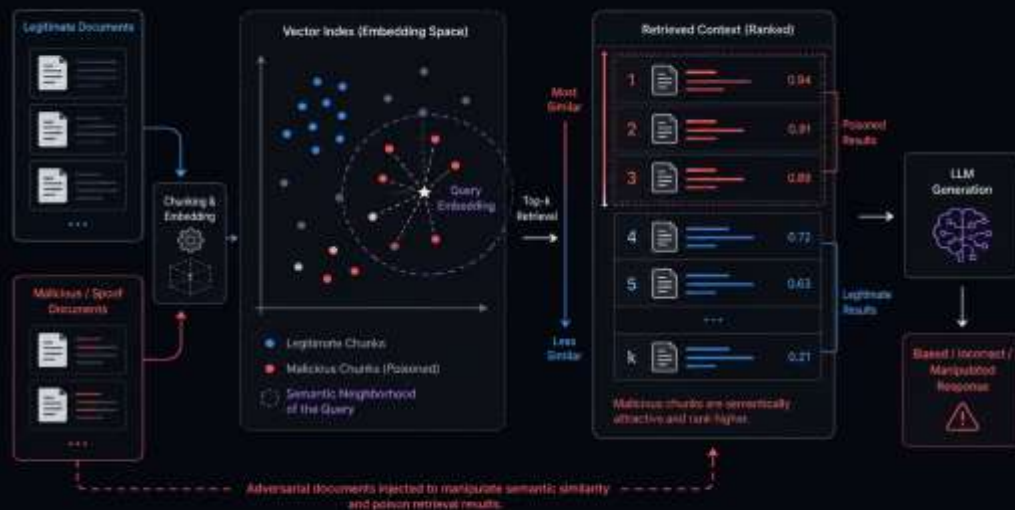
Research Novelty

Attack occurs entirely during retrieval
No prompt manipulation or jailbreak required
Query-aware retrieval robustness evaluation
Invisible to generation-time defenses

What We Built

Semantic retrieval poisoning benchmark
Spoof generation pipeline (gradient-optimized)
Doc2Query / Reverse QA answerability signal
Multi-run statistical validation

Example Attack



Experimental Pipeline



~5,000 clean corpus documents 300 attacked validation queries Recall@5, Recall@20, Top-1 Spoof Win Rate Mean $\pm$ standard error over 5 runs

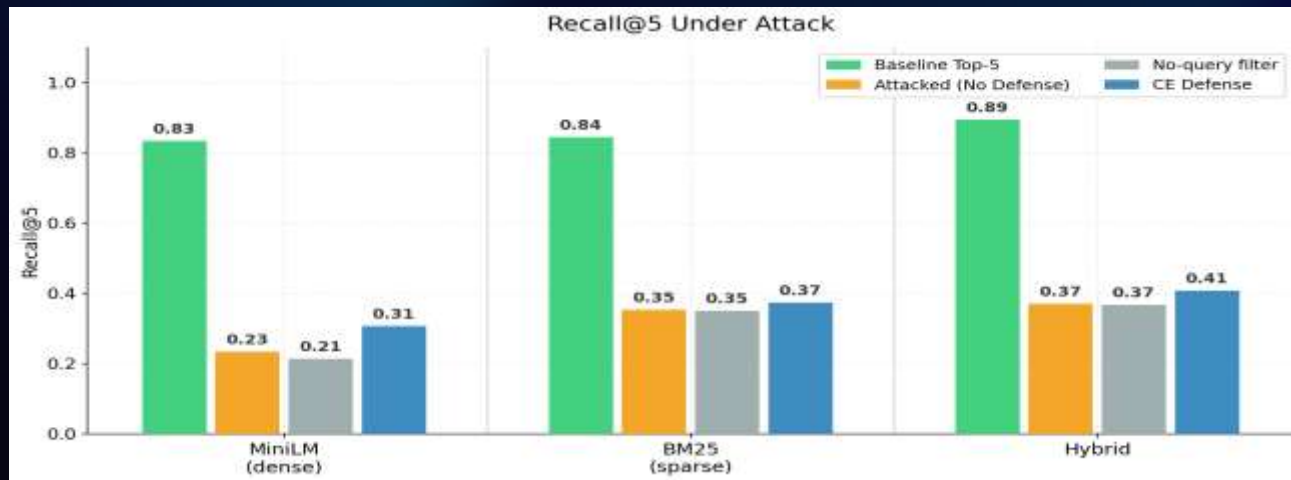
Core Technologies & Models

Technology / Model	Purpose	Short Explanation
SQuAD v1.1	Question Answering Dataset	Benchmark dataset used to build and evaluate the retrieval corpus.
FAISS	Vector Database	Performs efficient nearest-neighbor search over dense vector embeddings.
MiniLM (all-MiniLM-L6-v2)	Dense Retrieval	Converts queries and document chunks into embeddings for semantic similarity search.
BM25	Sparse Retrieval	Ranks documents using lexical matching based on TF-IDF-inspired scoring.
Hybrid Retrieval	Combined Retrieval	Fuses BM25 and MiniLM scores to improve retrieval robustness.
Cross-Encoder (ms-marco-MiniLM-L-12-v2)	Reranking	Re-evaluates the Top-20 retrieved chunks using joint query-document attention.
Reverse QA	Query-Aware Defense	Evaluates whether retrieved chunks can actually answer the user's question by generating hypothetical questions from each candidate chunk.
GPT-4o-mini	Question Generation	Generates hypothetical questions used by the Reverse QA defense.

Together, these components form a complete retrieval, attack, and defense pipeline for evaluating Retrieval Poisoning attacks in Retrieval-Augmented Generation (RAG) systems.

Retrieval Robustness Under Semantic Poisoning Attacks

Results averaged across 300 in the Top 5 attacked queries:



Key Findings

- Dense retrieval suffers the largest Recall@5 drop under attack
- BM25 retrieval is more robust against semantic spoofing
- Hybrid retrieval partially mitigates poisoning effects
- Query-aware defense provides only limited recovery

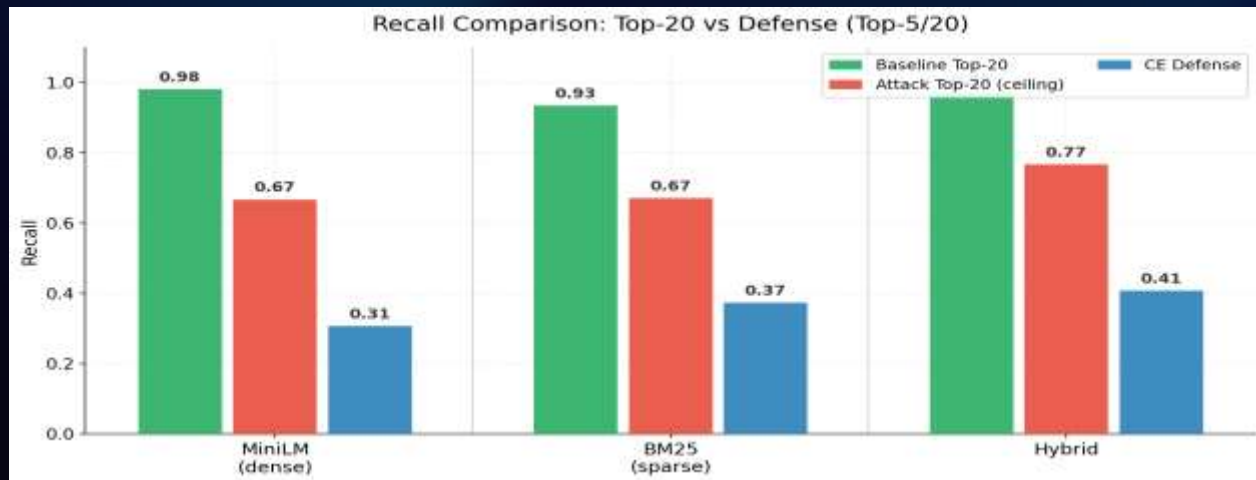
Final metrics to be populated with mean $\pm$ standard error across 1,000 validation queries



Final metrics to be populated with mean $\pm$ standard error across 1,000 validation queries.

Defense Performance is Bounded by Retrieval Recall

Results averaged across 300 in the Top 20 attacked queries :



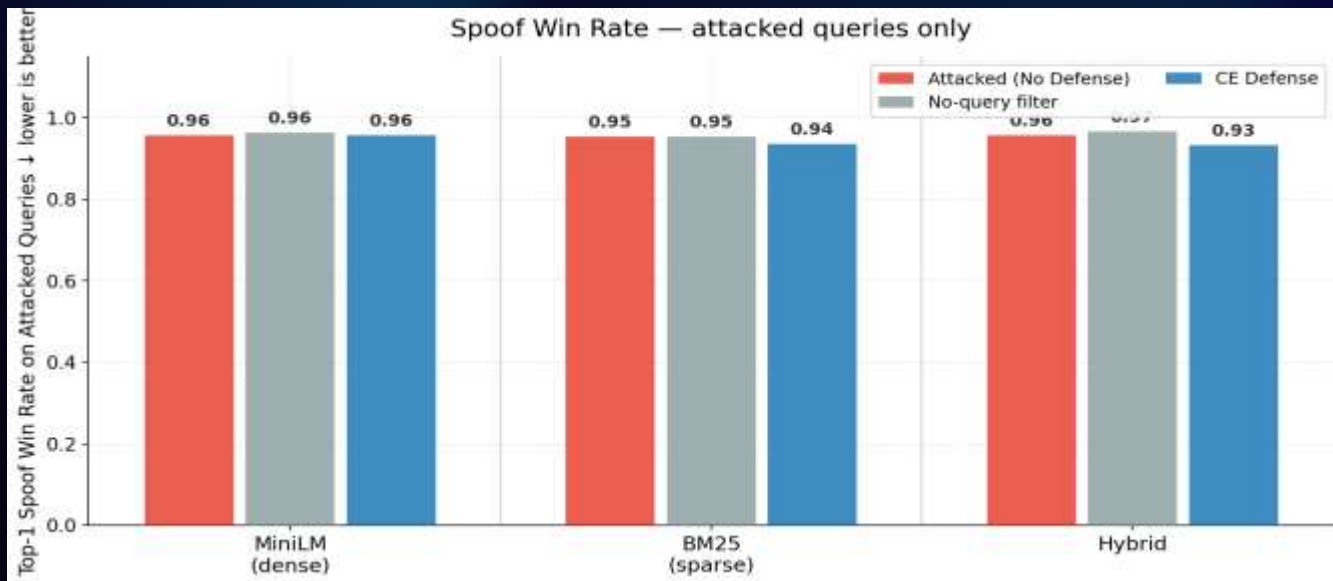
Key Findings

- The defense operates only on the Top-20 retrieved candidates.
- Therefore, its maximum achievable performance is bounded by Recall@20.
- If the relevant document is not retrieved, no reranking strategy can recover it.
- Stronger retrieval leads directly to stronger downstream defense.

❏ The conclusion: the defense cannot recover documents that were never retrieved.

Lightweight defenses provide only marginal improvement

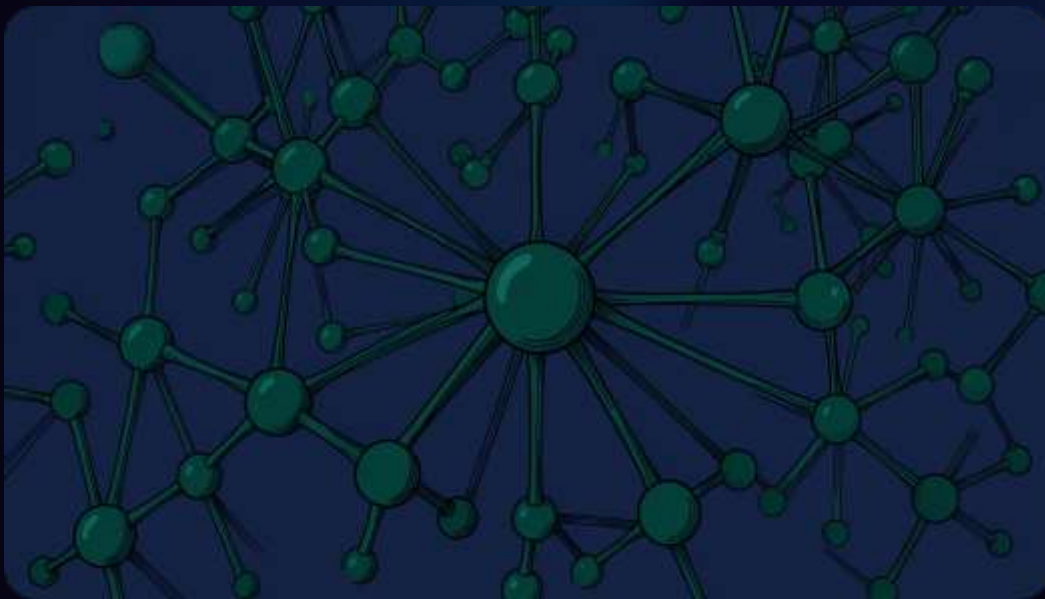
Top-1 Spoof Win Rate



Key Findings

- Dense retrieval shows the highest spoof success rate
- BM25 retrieval is slightly more robust under attack
- Hybrid retrieval partially reduces spoof dominance
- Lightweight defenses provide only marginal improvement

Conclusions & Future Work



Key Findings

Semantic retrieval poisoning is a practical, low-cost attack

Dense retrieval is highly vulnerable; sparse methods offer partial defense

Current RAG safeguards do not address corpus-level threats

Future Directions

Stronger document-level defenses against retrieval poisoning.

Advanced Reverse QA and answerability-based filtering.

Securing RAG systems requires protecting retrieval itself — not only generation.

Thank You

Questions?

We welcome discussion on retrieval robustness, adversarial NLP, and RAG security.

